- .NET has two parts: the .NET SDK (also known as the .NET CLI) and the .NET Runtime. When you're developing an application, you use the .NET CLI to restore, build, and run your application. Visual Studio uses the same .NET CLI commands from the IDE.
- When you want to deploy your app to production, you need to publish your application, using `dotnet publish`. This creates a folder containing your application as a DLL, along with all its dependencies.
- To run a published application, you don't need the .NET CLI because you won't be building the app. You need only the .NET Runtime to run a published app. You can run a published application using the `dotnet app.dll` command, where app.dll is the application .dll created by the `dotnet publish` command.
- To host ASP.NET Core applications in IIS, you must install the ASP.NET Core Module. This allows IIS to act as a reverse proxy for your ASP.NET Core app. You must also install the .NET Runtime and the ASP.NET Core Runtime, which are installed as part of the ASP.NET Core Windows Hosting Bundle.
- IIS can host ASP.NET Core applications using one of two modes: in-process and out-of-process. The out-of-process mode runs your application as a separate process, as is typical for most reverse proxies. The in-process mode runs your application as part of the IIS process. This has performance benefits, as no interprocess communication is required.